

MASTER STRATEGY DOCUMENT

Strategic Venture Plan

Mission, market analysis, competitive landscape, moat thesis, product strategy, capital plan, and the path from India to Asia to the world.

DOCUMENT VERSION

0.2

AUTHORED

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LAST MAJOR UPDATE

China + moat sharpening 2026-07-01

SINGLE POINT OF TRUTH

Yes — read this before refining anywhere

Status: Draft — pre-commit **Working title:** TBD (placeholder: **PadelMind** / **PadelGuru** / **CourtIQ**) **Authored:** 2026-07-01 **Authors:** Manoj Maheshwari + Claude (planning), padel coach TBD, ML co-founder TBD **Single Point of Truth:** This file. Update here before anywhere else. Memory pointer in `MEMORY.md`.

1. Mission & Vision

Mission

Make world-class padel coaching universally accessible — by turning every smartphone-equipped player into a data-driven student of the game, and every club into a smart venue.

Vision (10-year)

The default AI coaching infrastructure for padel in Asia. The system every serious player checks after a match the way runners check Strava and chess players check Lichess. A category-defining brand born in India that exports to the world — not the other way round.

What we will NOT be

- A recording-and-storage service (commoditized, no moat, no defensibility)
- A booking app (Hudle owns that lane in India; we partner, we don't compete)
- A scoreboard / scoring app (saturated, free alternatives)
- A B2C consumer-only app without club distribution (impossible CAC at our scale)

2. Strategic Objectives (3-Year Horizon)

HORIZON	OBJECTIVE	MEASURABLE TARGET
Year 1 (2026–27)	Validate product + 1 paying anchor city	25 clubs live in Mumbai/Pune/Bangalore; 5,000 unique players analysed; ₹50L MRR by Q4
Year 2 (2027–28)	Pan-India scale + first Asia foothold	200 clubs in India; 50,000 players; entry into UAE/Saudi via partnership; ₹3 Cr MRR; Series A closed
Year 3 (2028–29)	Asia-wide dominance + global pilots	1,000 clubs across India + UAE + KSA + SE Asia; 250,000 players; pilots in EU/LATAM; ₹15 Cr MRR; Series B ready

North Star Metric

Weekly active analysed players (WAAP). Not court installs, not bookings, not revenue. A player whose match has been analysed and who has opened the resulting feedback within 7 days. This is the only number that proves the moat is real.

3. Market Analysis

3.1 India (home market — beachhead)

METRIC	2023	2026 (NOW)	2036 PROJECTION
Padel courts	~80–100	~500	5,000+
Active players	<5,000	~100,000	1.5M+
Market size	~\$3M	\$25–30M	\$250–300M
New venues	12	48 in first 4 months alone	—
Growth rate	—	70% YoY annualised	—

Engagement signal: Games-per-player rose 73% YoY (5.3 → 9.2/year). Regular+heavy users = **34.5% of base** — highest of any racket sport on Hudle.

Capital signals: - Hudle closed **\$2.5M Series A in early 2025** — investors include **Mahesh Bhupathi** and **Ajinkya Rahane** - KheloMore: seed-stage, Clear Bridge Ventures - CAA Portas partnered with Hudle on India Padel Report 2026 — institutional money paying attention

Why India is the right beachhead for us: 1. Market is early-explosive (70% YoY) but still small enough that 25 clubs is meaningful penetration 2. No India-headquartered AI camera company exists — open category 3. English-medium, low-cost labour pool for labelling and BD 4. Founder credibility in India (existing AX network, WhatsApp distribution) 5. Bollywood / cricket adjacency = built-in influencer ecosystem (Bhupathi, Rahane already in the cap table of the booking layer)

3.2 Asia + Middle East (Year 2 expansion)

COUNTRY	COURTS	WHY IT MATTERS
UAE	950+	Largest single-country market in Asia. Affluent, expat-heavy, English-speaking. Easy first export
Saudi Arabia	900+ courts, 320 clubs	Vision 2030 sport-infra capex. Hosts Premier Padel. Projected \$162.8M domestic market by 2034. Sovereign-fund money
Indonesia, Thailand, Pakistan, China	Combined ~1,000+	Emerging, low-competition, lower ASP but volume play
Asia total	4,600+ courts, 1,700+ clubs across 30+ countries	Big enough to be a real second market

Massive catalyst: Padel ratified for the 2026 Asian Games (Nov 2025). Every Asian national federation now scrambling to form. National training centres about to receive government budget. **This is a once-per-sport-lifecycle window.** We must be visible to the new federations within 12 months.

3.3 Global (Year 3+)

- **58,300 courts globally** (2026), projected **91,000 by 2028**
- **19–25M active players** worldwide
- Europe = 55% of courts (saturated, competitive); LATAM = 23% (Argentina, Mexico — home of the sport)
- Search interest **+1,026%** since 2004 — fastest-growing sport on the planet
- Padel market projected **CAGR 12.93%** through 2035

We don't need to win globally. We need to win Asia (~5,000+ courts addressable). Global is bonus / acquisition surface.

4. Competitive Landscape

4.1 Direct AI-camera competitors (global)

PLAYER	HQ	FUNDING	CLUBS	STRENGTHS	WEAKNESSES
Clutch	Denmark / Lebanon	Pre-seed (small)	125+ worldwide	Best UX, gesture-trigger moat, mature highlights	No Asia footprint, expensive in INR (~€179/court/mo), no domain expert visible
Wingfield	Germany	Unknown (likely \$5–20M raised)	Several hundred	Best-known brand in EU, multi-sport (tennis/padel/pickleball)	Tennis-first DNA, EU-only, premium pricing
Padelytics	Spain	Partnered with SportAI	Europe/US/LATAM	Cloud-based AI, real-time deployed	Now dependent on SportAI partnership; no India presence
Eyes-On	Spain	Unknown	EU	Established brand in Spanish padel	Spanish-language-first; no Asia
Save My Play	USA	Integrated SportAI	US clubs	Strong US distribution	No padel-specific depth, no Asia
PlaySight (SmartCourt)	Israel	\$40M+ raised	Global	Multi-sport, well-funded	Expensive enterprise install, slow to ship; padel is a side-arm
GAMETRAQ / GameCam	UK	Unknown	UK/EU	Coach-focused	Niche, no scale
PB Vision	USA	Unknown	Pickleball-first	Backend used by other apps	Not padel-native

4.2 India-specific landscape

- **No India-headquartered AI camera competitor exists** (verified 2026-07-01)
- **Hudle** — booking + community, ~90% of India bookings, **complement not competitor**. Joint case for a partnership: their booking → our post-match analytics
- **KheloMore** — booking app, same pattern as Hudle, smaller scale
- **Padel India / Padel.in / regional brands** — operate courts, not tech players
- **Cricket-adjacent infra companies** (Decathlon, Sports Mechanics, etc.) — no overlap

4.3 Adjacent threats (could pivot in)

- **Hudle itself** could build analytics (low probability — capital-intensive, not their muscle)
- **SwingVision** (tennis app) could expand to padel (medium probability — well-funded, consumer DNA)
- **Epic Padel / Padel India / Zero.40** — Already revealed (see 4.2.1) — vertically integrated US-backed rollup, \$10M raised, India presence via Curved Consistency Technologies Pvt Ltd (Bangalore), co-founded by Sharath Kumar (ex-Ninjacart, ex-BharatPe)

4.2.1 The Indian Clutch operation — full disclosure (added 2026-07-01)

The Indian arm is **not a distributor — it's part of a vertically integrated global rollup**:

- **Indian entity**: Curved Consistency Technologies Private Limited
- **Brand**: The Clutch Sports (thecutchsports.com)
- **Office**: Indiranagar, Bangalore
- **Co-founder in India**: **Sharath Kumar** (ex-Ninjacart, ex-BharatPe, HBS Online exec ed)
- **Parent ecosystem**: Epic Padel (\$10M seed Sept 2025, led by NowaisWorld + Stryde Ventures) — also owns Padel India clubs, Red Padel (World Padel Rating), Pro Padel League, Padel Haus, Two Two
- **Imminent launch**: Zero.40 (2026) — “club operating system” combining Clutch AI + Red Padel rating + booking + community

Implication: what looked like “a Danish startup with no India presence” is actually a US-backed vertically integrated rollup whose Indian arm is a Bangalore-incorporated entity. They could lock up courts, players, ratings, AND camera tech in one stack. The window is months, not years.

Reach-out plan: LinkedIn DM to Sharath Kumar as a peer (AX founder + padel enthusiast) — see live message thread for draft.

4.4 China — the threat I missed in v0.1 (added 2026-07-01)

China padel is bigger, more state-backed, and producing competitors faster than initially briefed.

Market signals: - ~\$120M annual market today, projected top-5 globally by court density - **State-backed** youth pipeline: 12,000 athletes trained across 200 schools in 15 provinces since 2023 (Chinese Tennis Association initiative) - Youth tournament participation +300% YoY - Beijing National Tennis Center → 8 padel courts; Shanghai Xuhui → 6 glass-court flagship

Chinese competitors — ranked by threat horizon:

THREAT	WHO	WHY	TIME TO PADEL
HIGH (12 months)	(Xiaoqiuquan)	ex-ByteDance + ex-DJI founders. Already AI-coaches tennis/badminton/table-tennis/ pickleball . Phone-side light model + cloud large-model arch. Top-50 China sports app. ~30% MoM user growth.	Adding padel = config change
MEDIUM (18 months)	Lumistar (Shenzhen)	TERO tennis AI training system launched Mar 2026. 4K dual-cam + sensor array.	Cloud-analytics extension
MEDIUM (variable)	SenseTime	Already does AI analytics for Chinese pro basketball, 3D motion capture for table tennis/archery.	Sponsored entry
LOW (giants)	Alibaba Cloud	Olympics bullet-time multi-cam replays. Won't enter directly but white-labels to whoever wins China.	Infra-only
LATENT	Hikvision, Dahua, Reolink, Uniview	They make the cameras every padel AI startup buys. Vertically integrated padel play is technically trivial for them.	Strategic choice not capability

4.5 Reality check — what are the real moats? (added 2026-07-01)

Pre-China analysis was too optimistic about tech as a moat. Hard truth:

MOATS THAT DO NOT WORK against capable Chinese/global competition: - ❌ AI quality (commodity ML pieces glued together; CV talent is global) - ❌ Hardware (Chinese ODMs make the cameras anyway) - ❌ Pricing (Chinese players can match or beat us at any tier) - ❌ Dataset size (volume can be matched within 12 months by anyone funded)

MOATS THAT DO WORK:

MOAT	STRENGTH	WHY
India geopolitical wall	HIGH	India app bans (TikTok 2020, PUBG, WeChat, SHEIN-class) extend to Chinese sports SaaS. Indian clubs face procurement risk choosing Chinese tech. This is our shield in India.
Federation distribution lock	HIGH (if won early)	Asian Games 2026 catalysed national padel federations forming NOW (India, UAE, KSA, SE Asia). Lock procurement at federation level → blocks Chinese entry by policy.
Coach co-founder + domain rubric	MEDIUM	A WPT-trained-coach-led rubric is hard to replicate without their own sport credibility hire. Buys 12–18 months.
WhatsApp-native distribution + Hindi/Arabic UX depth	MEDIUM	AX-style cultural-distribution muscle Chinese SaaS rarely matches.

Strategic question this raises (logged in Open Questions): Should we deliberately be “India + Middle East only”, explicitly anti-China by geography, and partner-or-license China rather than compete? Defending one geographic moat well > losing across three.

4.6 China watch list — monthly monitoring (added 2026-07-01)

Add to ops cadence — review monthly, escalate any padel-pivot signal within 24 hrs:

TARGET	WHAT TO WATCH	SIGNAL THRESHOLD
(Xiaoqiuquan)	App store listings, blog posts, partnerships	“Padel” mentioned anywhere in product copy
Lumistar	Press releases, CES, racket-sport announcements	Padel hardware SKU
SenseTime sports	Pro team partnerships, demo videos	Padel federation deal
Hikvision / Dahua	Vertical product launches	Padel-specific SKU or partnership
Padel-Magazine, Wamda, Sports Tech feeds	Funding announcements, China deals	Any China-padel-tech raise > \$5M

4.4 Our positioning vs incumbents

DIMENSION	CLUTCH / WINGFIELD	US
Geography	EU-first	Asia-first, India-native
Pricing	€179/court/mo (~₹17,000)	₹3,000–7,000/court/mo (Indian SMB-priced)
Domain expertise	Tech team only	Sport-expert co-founder (coach)
Language	EN, ES, IT	EN + Hindi + (later) Arabic, Bahasa, Mandarin
Feedback model	Highlights + stats	Coaching feedback (“drill X for 5 min”)
Distribution	Direct sales	Hudle partnership + WhatsApp-native funnel
Hardware cost	Hidden (bundled)	Transparent ₹15–20k camera + ₹500/mo (camera optional after Y2)

Core differentiator: We are an AI padel COACH, not an AI padel CAMERA. That framing wins on every dimension above.

5. Product Strategy

5.1 Three product surfaces — sequenced

Surface 1: Club System (Year 1 focus, B2B2C) - IP camera + edge buffer + cloud AI per court - Club dashboard: court usage, recording index, sponsor overlay tools, player engagement - Player PWA: post-match feedback, shot histogram, skill score, drill recommendations - **Pricing:** ₹3,000–7,000/court/mo subscription; hardware ₹15–20k upfront or rolled into 24-mo plan

Surface 2: Coach Tool (Year 2, B2C2B) - Coach-facing dashboard: review labelled clips from their students; assign drills; track progress over months - **The lock-in.** Once a coach has 30 students on the platform, their workflow is on us - **Pricing:** ₹500–1,500/coach/mo, or free if they bring N students (acquisition lever)

Surface 3: Player Skill Identity (Year 2–3, B2C) - The “Clutch Score” equivalent that follows a player across clubs and cities - Cross-club leaderboards; skill-matched matchmaking; tournament seeding feed - **The network effect.** Hudle has bookings; we have *skill identity* — these are complementary, not competing - **Pricing:** Freemium. Free tier = score + 5 analyses/mo. Pro = ₹399/mo, unlimited

5.2 What we build in V1 (next 12 weeks — locked)

COMPONENT	STATUS	OWNER
Rubric document (shot taxonomy + form-quality criteria)	TBD	Coach
Camera install at 1 pilot court	TBD	Manoj
<code>padel_analytics</code> fork — streaming refactor	TBD	ML lead
Shot classifier (V1, ~70% accuracy target)	TBD	ML lead
Player PWA (mobile-first React)	TBD	Manoj + AI pair
Club dashboard (basic)	TBD	Manoj + AI pair
Highlight auto-clipper	TBD	Manoj + AI pair
Skill score calc engine	TBD	Manoj + AI pair

5.3 What we DO NOT build in V1

- Native Android/iOS apps (PWA is enough; saves 8 weeks)
- Live-streaming integration (Year 2)
- Gesture-trigger (“save the last point” hand wave) — clever, but Clutch’s moat, copy in Y2
- Sponsorship overlay tools — Year 2
- Multi-camera / 3D ball depth — Year 3 (Series B feature)
- Line-calling / officiating — explicitly not in scope; high failure cost

6. Technical Architecture & Feasibility

6.1 The stack (validated by `padel_analytics` audit done 2026-07-01)

```
[IP Camera @ court]
| RTSP H.265
▼
[Edge Buffer – Raspberry Pi 5 + 1TB NVMe]
| Match-end trigger uploads MP4
▼
[Cloudflare R2 – match storage]
|
▼
[RunPod / Lambda GPU worker]
- YOLOv8m player detection
- YOLOv8-pose keypoints
- TrackNet ball tracking
- Court homography (from 12 fixed keypoints clicked at install)
- Shot classifier (our trained model) ← THE MOAT
- Skill scoring engine ← THE MOAT
|
▼
[Supabase: structured analytics + player identity]
|
▼
[Player PWA – Cloudflare Pages] + [Club dashboard]
|
▼
[WhatsApp delivery – AX router or Twilio]
```

6.2 Feasibility verdict (per repo audit)

LAYER	RISK	VERDICT
Camera + RTSP	Low	Off-the-shelf Reolink RLC-823A handles 4K RTSP. Validated by hobbyist sports projects
Court homography	Low	<code>padel_analytics</code> already supports click-once-cache-forever flow
Player tracking	Low	YOLOv8m + ByteTrack is a solved problem
Pose keypoints	Low	YOLOv8-pose is mature
Ball tracking	Medium	TrackNet works but needs retraining on padel-specific data (ball lighter colour vs tennis)
Shot classification	HIGH	No OSS solution exists. Must train. ~70% V1 target. This is the project's central technical risk
Skill scoring	Medium	Composite of pose-quality + outcome + context. Algorithmic, not ML, but tricky to tune to coach intuition
Streaming refactor of <code>padel_analytics</code>	Medium	Batch-style today; ~3 weeks of work to make single-pass-per-frame
Mobile PWA	Low	Standard React work

6.3 Hardware bill of materials — per court

ITEM	SPEC	INR
Camera	Reolink RLC-823A 16x (4K, PoE, IP66) — or Hikvision premium variant for 50% more	12,000–22,000
PoE switch	TP-Link TL-SG1005P	3,000
Edge buffer	Raspberry Pi 5 8GB + 1TB NVMe + case	15,000
Cabling + mount	Generic	2,000
Total per court		~₹32,000–42,000

6.4 Cloud cost — per court per month (at 120 matches/court/mo)

ITEM	INR
R2 storage + egress	~150
RunPod GPU inference (~3 min/match × ₹2.5/min)	~360
Supabase free tier	0
WhatsApp delivery (AX router)	~30
Total	~₹540/court/mo COGS

At ₹5,000/court/mo ASP → **~89% gross margin**. Healthy SaaS economics. Same as Clutch's underlying unit economics.

7. Team & Organisation

7.1 Founding team (target structure)

ROLE	PERSON	EQUITY	CASH
CEO / Product / Capital	Manoj	40–50%	Founder salary deferred
CTO / ML / CV	João Silva (target) OR equivalent	20–30%	Modest founder salary
Chief Coach / Domain (CCO)	TBD (named coach)	8–15%	₹50–80k/mo or equity-heavy
Founding Engineer	Hire month 6	1–2%	₹12–18 LPA
Founding BD	Hire month 9	0.5–1%	₹10–15 LPA + commission

ESOP pool: **15% reserved**. Total founder + employee dilution at pre-seed: ~85% founders + 15% ESOP.

7.2 Manpower plan by stage

STAGE	HEADCOUNT	ROLES
Pre-seed (months 0–6)	3 (founders)	CEO + CTO + CCO. Contractors for labelling (₹15k/mo × 2)
Seed (months 6–18)	8	+ 2 ML engineers, 1 hardware ops/install lead, 1 BD, 1 customer success
Series A (months 18–36)	25	+ Asia BD lead, regional install teams (3), more ML, design, content/coaching team (2)

7.3 Advisory board (target)

- 1× racket-sports business veteran (Bhupathi / Hudle adjacent — feasible)
- 1× CV/ML academic (UPC-ViRVIG team, or IIT prof)
- 1× former Premier Padel player (sport credibility)
- 1× venue/franchise operator (real-estate / club economics)

Advisor equity pool: 1–2% total, 0.25–0.5% each, 2-yr vest.

8. Infrastructure Requirements

8.1 Compute

- **Year 1:** RunPod / Lambda on-demand. ~₹50k/mo at 25 clubs × 120 matches each
- **Year 2:** Reserved GPU instances on AWS/GCP. ~₹5L/mo at 200 clubs
- **Year 3:** Dedicated GPU cluster (~10× A100s) hybrid cloud. ~₹25L/mo at 1,000 clubs

8.2 Storage

- Cloudflare R2 throughout (no egress fees vs AWS S3 saves ~60%)
- Cold-tier matches >90 days to keep costs down
- Year 3 estimate: ~200 TB hot + 1 PB cold

8.3 Software infra

- Supabase (Postgres + Auth + Storage + Realtime) — same stack as AX, leverage existing operator muscle
- Cloudflare Pages + Workers for PWA + edge logic
- Sentry, PostHog for observability
- GitHub Actions for CI/CD
- Linear for product tracking, Notion for docs

8.4 Hardware logistics

- **Year 1:** Manual install per court. ~₹5k labour per install
- **Year 2:** Trained install partner network (3 cities)
- **Year 3:** Self-install kit + remote-assisted (target ₹1k labour cost)

9. Capital Plan

9.1 Funding stages

ROUND	WHEN	AMOUNT	USE	INVESTORS TO TARGET
Bootstrap / Founder cash	Now (months 0–3)	₹15–25L (founder + close angels)	1 pilot club, V1 build, rubric IP, MVP video	Self + 2–3 angels
Pre-seed	Month 4–6	₹2–3 Cr (\$250–350k)	5 clubs live, founding team salaries 12 mo, hardware for 25 clubs	Sky Impact (Hudle's lead), Stellaris, Blume Founders Fund, Better Capital, sports-angel syndicates
Seed	Month 12–15	₹15–20 Cr (\$1.8–2.5M)	India 200-club scale, UAE entry, ML team build, Series A prep	Accel, Lightspeed India, Matrix, Elevation, regional SWFs (UAE)
Series A	Month 30–36	₹80–120 Cr (\$10–15M)	Pan-Asia scale, global pilots	Sequoia/Peak XV, Tiger, regional sovereign funds

9.2 Cash burn forecast (₹ Lakhs/month)

MONTH	HEADCOUNT	BURN	REVENUE	NET
1–6	3 + contractors	4	0	-4
7–12	5	12	2	-10
13–18	10	25	15	-10
19–24	14	40	50	+10
25–36	25	90	180	+90

Break-even target: Month 22–24. Series A raised before breakeven for Asia expansion fuel, not survival.

9.3 Capital efficiency moats we should preserve

1. **AX shared infra reuse** (Supabase patterns, WhatsApp router, dashboards) saves ~6 months of platform work
 2. **India ML/labelling cost** is 1/5th of EU equivalents
 3. **Cloud R2 vs S3** saves 50–60% on video storage
 4. **PWA over native** saves 8 weeks + ongoing native team
-

10. Go-to-Market Plan

10.1 Year 1 — 3 cities, 25 clubs, 5,000 players

Beachhead: Mumbai → Pune → Bangalore. Why: highest court density, highest disposable income, strongest WhatsApp + tech-adoption signals.

Acquisition wedge — “Free pilot for top 3 clubs per city”: - We install hardware + run analytics free for 60 days - Club brings 30+ regular players who try the system - After 60 days: club pays subscription OR we remove hardware (low-stakes for them) - Players who tasted the feedback don’t want to lose it → club gets pressured by demand - This is the same playbook Hudle used to seed its booking dominance

Distribution levers: 1. **Hudle partnership** — explicit ask: in-app push to players in clubs we’ve covered, in exchange for revenue share or data 2. **WhatsApp-native funnel** — player gets feedback link via WhatsApp post-match; one tap to view, share, save → classic AX-style funnel 3. **Coach affiliate program** — coaches earn 10% recurring on every player they refer; this creates a sales force we don’t pay salary for 4. **Tournament sponsorships** — sponsor 1–2 local circuits, get sole AI camera + livestream rights, build brand

10.2 Year 2 — UAE entry

Why UAE first (not KSA): - English-default operating language - High-density (950+ courts in a small geography) - Expat-heavy player base = high willingness to pay - Easier business setup than KSA - Bridge to KSA via DIFC/ADGM-based operations

Entry approach: Acquire or partner with 1 existing UAE court operator chain (think 20-court operator) as launch customer. We become their tech stack, they become our distribution.

10.3 Year 3 — KSA + SE Asia

- KSA via Vision-2030 sport-infra RFPs; bid through UAE-registered subsidiary
 - SE Asia (Indonesia, Thailand) via partnership with local franchise operators
 - EU/LATAM pilots only if explicitly requested by investors (no own-cash spend)
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11. Phased Roadmap

Year 1 — Validate + India anchor

Q3 2026: V1 ship, pilot club live, 200 matches analysed, rubric IP locked, pre-seed raised **Q4 2026:** 5 clubs live, 1,000 matches/mo, first paying clubs, founding team complete **Q1 2027:** 15 clubs live, ₹15L MRR, seed conversations begin **Q2 2027:** 25 clubs live, ₹50L MRR, seed closed

Year 2 — Scale India + UAE entry

Q3 2027: 75 clubs, hardware ops team in place, UAE feasibility study **Q4 2027:** 125 clubs, UAE pilot launched (5 clubs) **Q1 2028:** 175 clubs in India, 15 in UAE, ₹2 Cr MRR **Q2 2028:** 200 + 30, ₹3 Cr MRR, Series A closed

Year 3 — Asia dominance + global pilots

Q3 2028: KSA entry (10 clubs), SE Asia partnerships signed **Q4 2028:** 400 India, 60 UAE, 25 KSA, 30 SE Asia → 515 clubs **Q1 2029:** EU pilot (5 clubs), LATAM pilot (5 clubs) **Q2 2029:** 750+ clubs, ₹10 Cr MRR, Series B raised or strategic acquisition conversations

12. Bottlenecks, Risks & Failure Modes (honest)

12.1 Technical bottlenecks

RISK	LIKELIHOOD	SEVERITY	MITIGATION
Shot classifier plateaus below 70% accuracy	Medium	High (kills product)	Multi-coach labels, larger dataset, ensemble models, hire ML specialist if stuck at month 8
Ball tracking unreliable in low light	Medium	Medium	Better camera (Hikvision ColorVu), club-side lighting recommendations, manual rally-end fallback
Cloud inference costs spike	Low	Medium	Edge inference V2 (Jetson) by Y2
<code>padel_analytics</code> upstream abandoned	Low	Medium	Fork early, don't depend on upstream

12.2 Market bottlenecks

RISK	LIKELIHOOD	SEVERITY	MITIGATION
Hudle decides to build this in-house	Medium	Very High	Partner formally before Hudle thinks about it. Lock revenue-share and data-share contract
Clutch enters India aggressively	Medium	High	First-mover speed advantage. 25 clubs by Q4 = defensible local network
Padel growth in India stalls (only 70% YoY for 1 year vs 5)	Low	High	Bet is on the growth continuing through 2028 Asian Games tailwind
Asian Games 2026 padel underwhelms public interest	Low	Medium	India has other catalysts (Bhupathi push, IPL-style league rumours)
Players unwilling to pay ₹399/mo Pro tier	Medium	Medium	Lower to ₹199, lean on club-paid subscription as primary revenue

12.3 Team bottlenecks

RISK	LIKELIHOOD	SEVERITY	MITIGATION
No ML co-founder lands	High	Very High	Two tracks: João email + active search via IIT/IISc network. 2-month cutoff before falling back to senior contractor
Coach overcommits, under-delivers	Medium	High	Clear deliverables doc + payment milestones, not pure equity
Manoj overstretched (AX is day job)	High	High	Honest weekly hour budget; raise pre-seed early enough to afford a founding engineer who removes the bottleneck
Hardware install team can't scale	Medium	Medium	Train partner network from Year 1 (don't centralize)

12.4 Capital bottlenecks

RISK	LIKELIHOOD	SEVERITY	MITIGATION
Pre-seed doesn't close by month 6	Medium	High	Have an SAFE-bridge of ₹50L from 3–5 angels lined up; pitch on Hudle thesis traction
India sports-tech FOMO cycle ends	Low	Medium	Padel is currently the hottest category — even with macro winter, this lane is funded
₹/USD FX deterioration affects hardware costs	Medium	Low	Negotiate INR contracts with Reolink India distributor; lock 12-mo BOM pricing

12.5 Failure modes that could kill the venture (existential)

- 1. Coach quality of judgment doesn't translate to algorithm.** The classifier hits 85% accuracy but real players say "this is wrong, I'm not playing badly" — losing trust kills WoM. *Mitigation: invest heavily in human-in-the-loop feedback before scaling.*
- 2. Hudle builds it or partners with Clutch directly.** Distribution moat collapses. *Mitigation: get to Hudle BEFORE Clutch does. They are in Mumbai.*
- 3. Founder burnout — Manoj cannot sustain AX + this.** *Mitigation: be honest about bandwidth; raise pre-seed quickly so a founding engineer can take ops load.*
- 4. Hardware reliability nightmare.** Cameras failing in monsoon, dust, summer heat → support ticket avalanche. *Mitigation: spend Y1 obsessing over hardware QA; standardize one camera model + thermal-tested install kits.*
- 5. Players don't actually share/talk about the feedback.** No viral loop → CAC stays high → unit economics break. *Mitigation: design feedback to be screenshot-worthy and WhatsApp-shareable from day one. Vanity-friendly visuals.*

13. Decision Gates (when to kill, when to push)

GATE	WHEN	PASS CRITERIA	IF FAIL
Gate 0 — Partnership	Week 1	Coach committed in writing + first club access confirmed	Pause venture, revisit in 60 days
Gate 1 — Rubric ships	Week 4	15-page coaching rubric doc complete and validated by 2nd coach	Replace coach
Gate 2 — V1 inference works	Week 8	End-to-end pipeline runs on 5 test matches; outputs make sense to coach	Pivot to contractor on classifier
Gate 3 — Player feedback validates	Week 12	≥6 of 10 testers say "I'd pay ₹300/mo for this"	Pivot product (coach tool? video reels?)
Gate 4 — Pre-seed term sheet	Month 6	₹2 Cr+ term sheet from a credible lead	Bootstrap 6 more months on bridge cash
Gate 5 — Hudle/distribution deal	Month 9	Formal partnership or proven independent CAC	Re-strategize GTM
Gate 6 — 25 clubs ARR	Month 15	₹50L MRR run rate	Pause UAE, double down India

14. The 30-Day Punch List (what we actually do this month)

1. **Week 1 — Partner lock-down** - [] Send João Silva collaboration email - [] Onboarding conversation with coach (define scope + comp) - [] Visit / call 3 clubs in Mumbai or Pune; pick pilot court - [] Register placeholder domain + form LLP / private limited paperwork
2. **Week 2 — Rubric + plan locked** - [] Coach delivers shot taxonomy v1 + scoring criteria v1 - [] Pilot club agreement signed (informal MOU OK) - [] Order hardware for pilot court (~₹35k spend) - [] Pre-seed deck v0 — Hudle thesis, Asian Games catalyst, team
3. **Week 3 — Install + first recordings** - [] Mount camera, fix the 12 court keypoints - [] First match recorded, uploaded, processed through `padel_analytics` - [] Coach reviews raw player+ball tracking, gives feedback - [] Begin recruiting ML co-founder (LinkedIn, IIT network)
4. **Week 4 — Rubric live + labelling begins** - [] Coach labels 100 gold-standard clips - [] 2 college labellers onboarded at ₹15k/mo each - [] Streamlit labeller tool built - [] First 3 angel conversations opened

Strategic Geographic Decision — pending (added 2026-07-01)

Given China findings, pick one before raising pre-seed:

- (a) **Pan-Asia ambition**: India → UAE → KSA → SE Asia → eventual China-via-partner. Bigger TAM, bigger raise, more competitive surface, China-collision risk in Year 3.
- (b) **"Anti-China by geography"**: India + Middle East only — explicitly defended by procurement-policy and language moats. Smaller TAM, defensible margin, simpler narrative. China handled via license/partnership/exit.

Manoj's lean: TBD. Decision impacts Section 9 (Capital Plan) raise size and Section 10 GTM.

Open Questions / TBDs (must resolve before raising)

- [] **Working name + domain** (current placeholders are weak; need brand)
- [] **Coach identity + commitment terms** (named, signed, on board)
- [] **ML co-founder identity** (João Silva or alternative)
- [] **Pilot club identity** (named, MoU)
- [] **Entity setup** — LLP, Pvt Ltd, or DIFC for future Asia? Tax & cap-table implications
- [] **Hudle relationship status** — warm intro, formal pitch, or compete?
- [] **Manoj weekly hour commitment** — honest 15 hrs/wk floor

Change Log

DATE	AUTHOR	CHANGE
2026-07-01	Manoj + Claude	v0.1 — initial draft
2026-07-01	Claude	v0.2 — Added Section 4.2.1 Clutch India full profile (Sharath Kumar / Curved Consistency / Epic Padel rollup), Section 4.4 China competitive landscape (Xiaoqiuquan, Lumistar, SenseTime), Section 4.5 moat reality check (tech moat downgraded; geopolitical + federation moats upgraded), Section 4.6 monthly China watch list, Strategic Geographic Decision pending

This is a living document. Update here before anywhere else. Memory pointer: `MEMORY.md` → `project_padel_clone_venture`.

Companion document set: Strategic Venture Plan · Project Report · Technical Feasibility · Commercial Report · Base Club Offer · Technical Review Request.

Single source of truth maintained in `~/Documents/padel-clone/`.